

Mathematical Induction

5.1 & 5.3

Chapter Summary

1. Mathematical Induction
2. Examples of Proof by Mathematical Induction
3. Recursive Definitions

Climbing an Infinite Ladder

Suppose we have an infinite ladder:

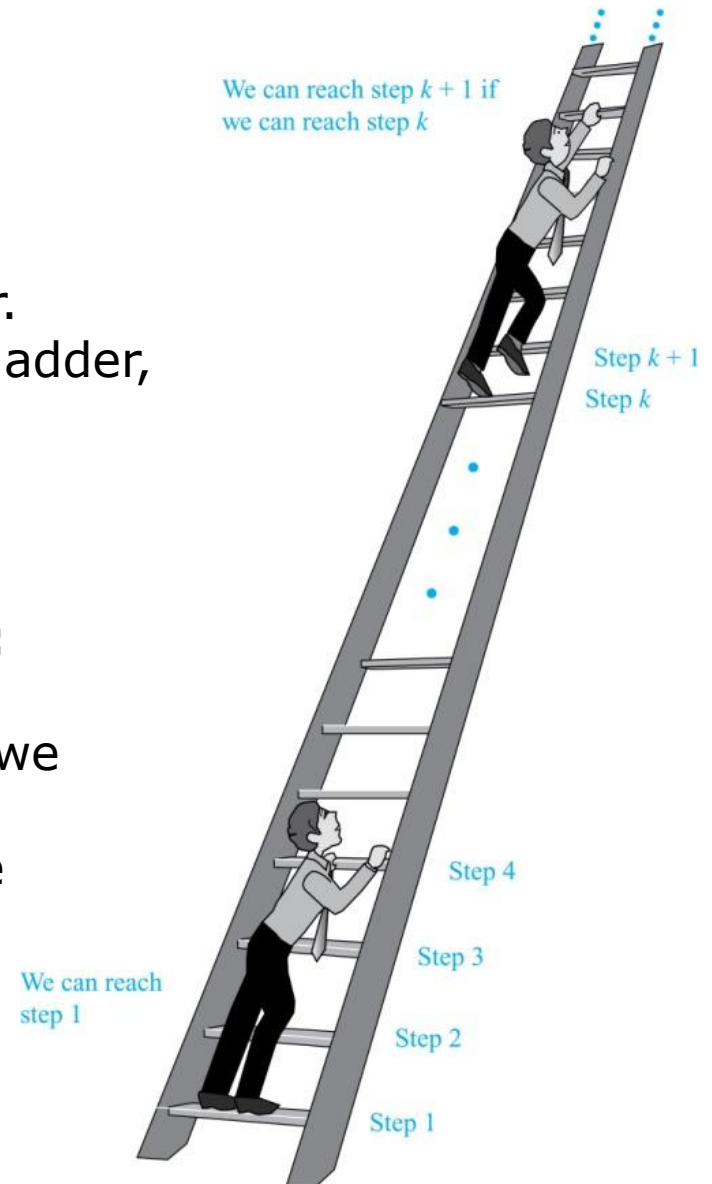
1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

The principle of Mathematic induction:

1. Show we can reach the first step
2. Show that if we can reach step k , then we can reach step $k+1$

if we establish these 2 things, we know we can reach all the steps of the ladder.

This example motivates proof by mathematical induction.



Principle of Mathematical Induction (MI)

Principle of Mathematical Induction: To prove that $P(n)$ is **true** for all **positive integers** n , where $P(n)$ is a propositional function, we complete these steps:

- 1. Basis Step:** Show that $P(1)$ is **true**.
- 2. Inductive Step:** Show that $P(k) \rightarrow P(k + 1)$ is **true** for all positive integers k .

Mathematical induction can be expressed as the rule of inference

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n P(n),$$

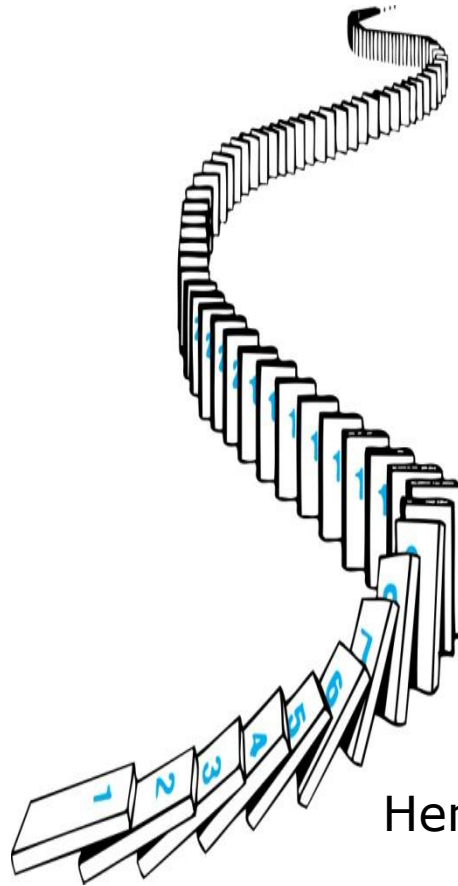
where the domain is the set of positive integers.

Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer. We will see examples of this soon.

Remembering How Mathematical Induction Works - Domino Effect

Consider an infinite sequence of dominoes, labeled $1, 2, 3, \dots$, where each domino is standing.

Let $P(n)$ be the proposition that the n th domino is knocked over.



We know that the first domino is knocked down, i.e., $P(1)$ is **true**.

We also know that if whenever the k th domino is knocked over, it knocks over the $(k + 1)$ st domino, i.e., $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

Hence, all dominos are knocked over.

$P(n)$ is **true** for all positive integers n .

Mathematical Induction

Suppose we have a sequence of propositions which we would like to prove:

$P(0), P(1), P(2), P(3), P(4), \dots P(n), \dots$

EG: $P(n) =$

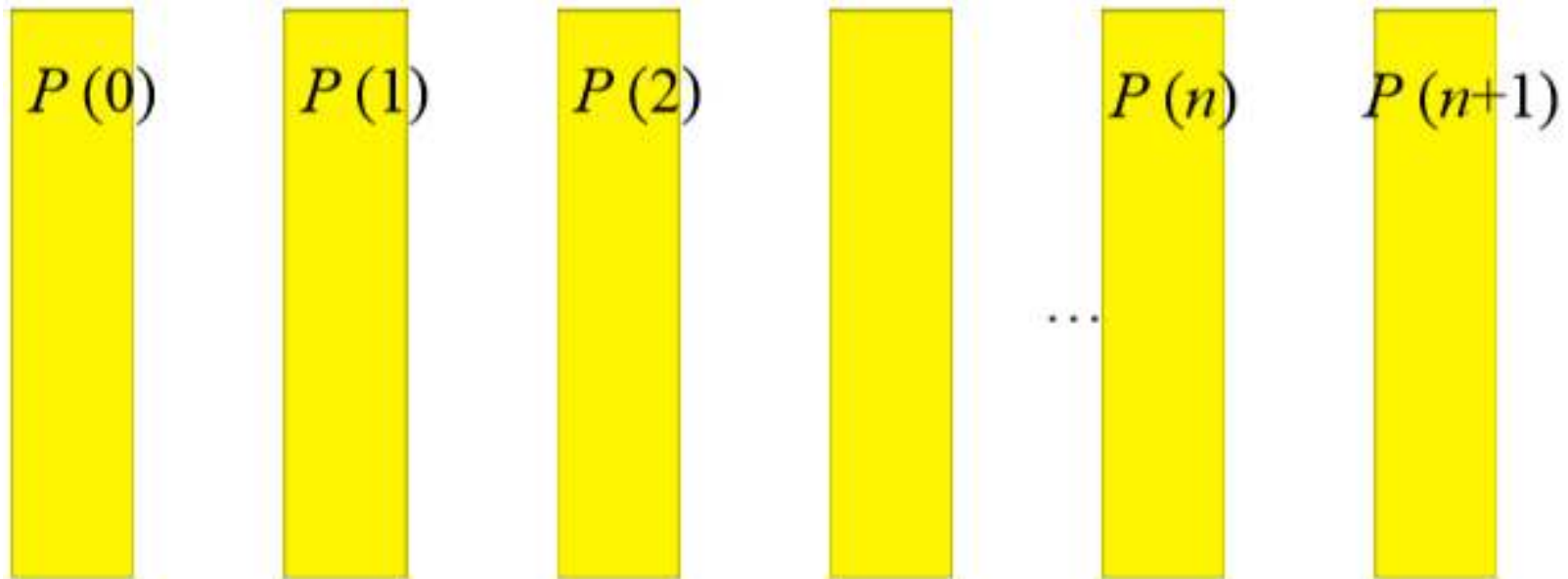
We can picture each proposition as a domino:



$P(n)$

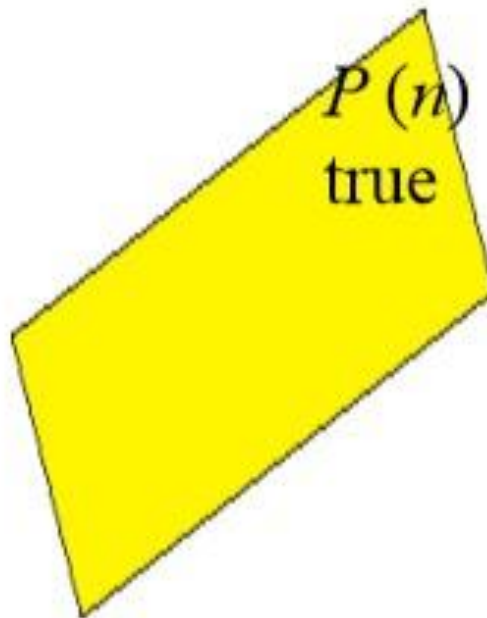
Mathematical Induction

So sequence of propositions is
a sequence of dominos.



Mathematical Induction

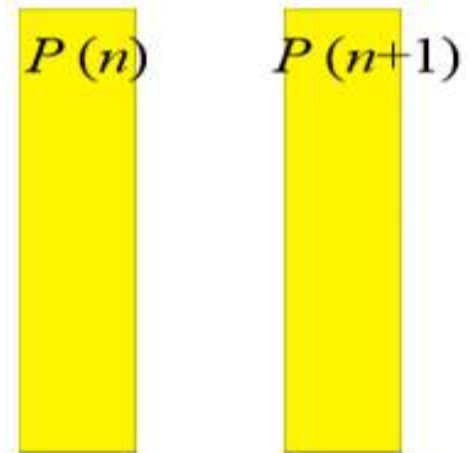
When the domino falls (to right), the corresponding proposition is considered true:



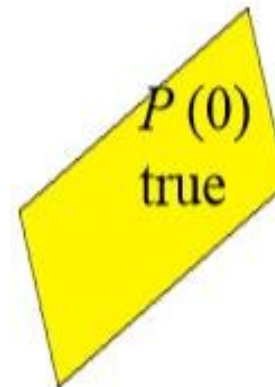
Mathematical Induction

Suppose that the dominos satisfy two constraints.

- 1) Well-positioned: If any domino falls (to right), next domino (to right) must fall also.

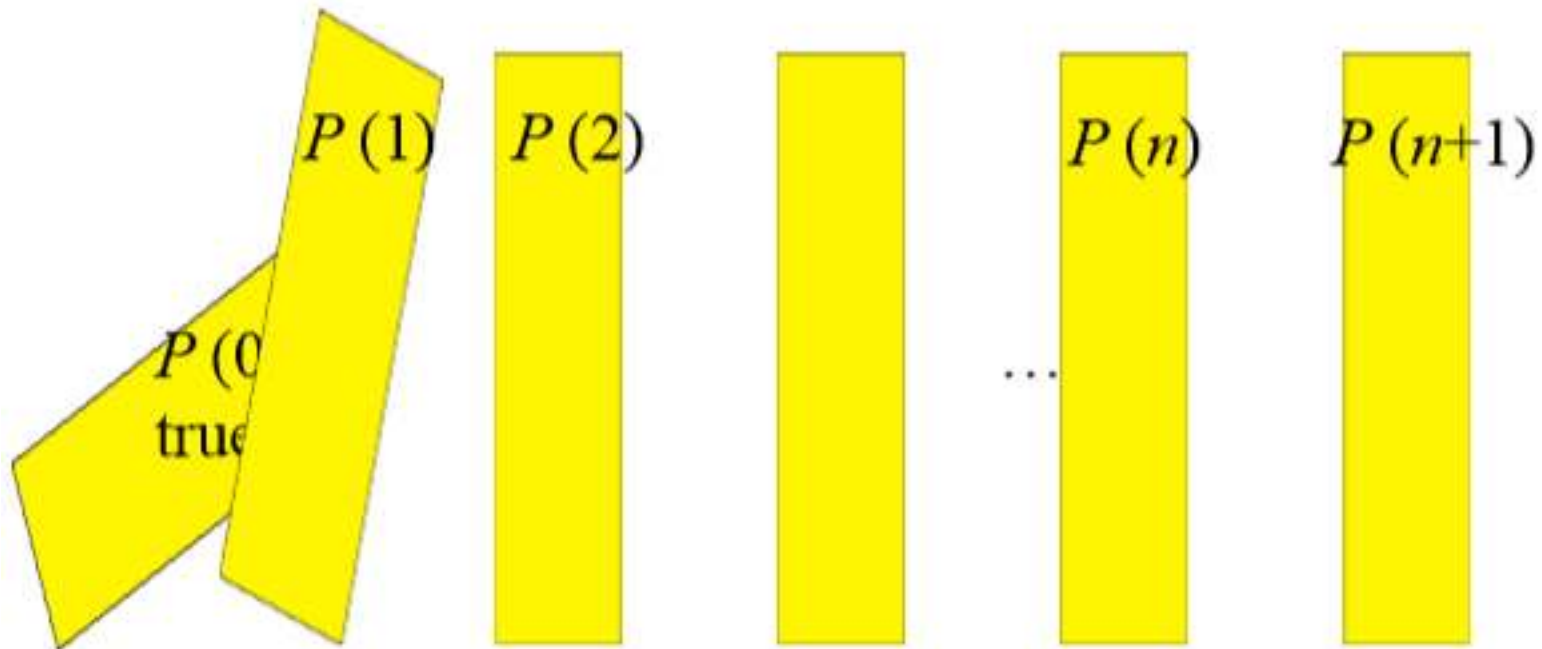


- 2) First domino has fallen to right



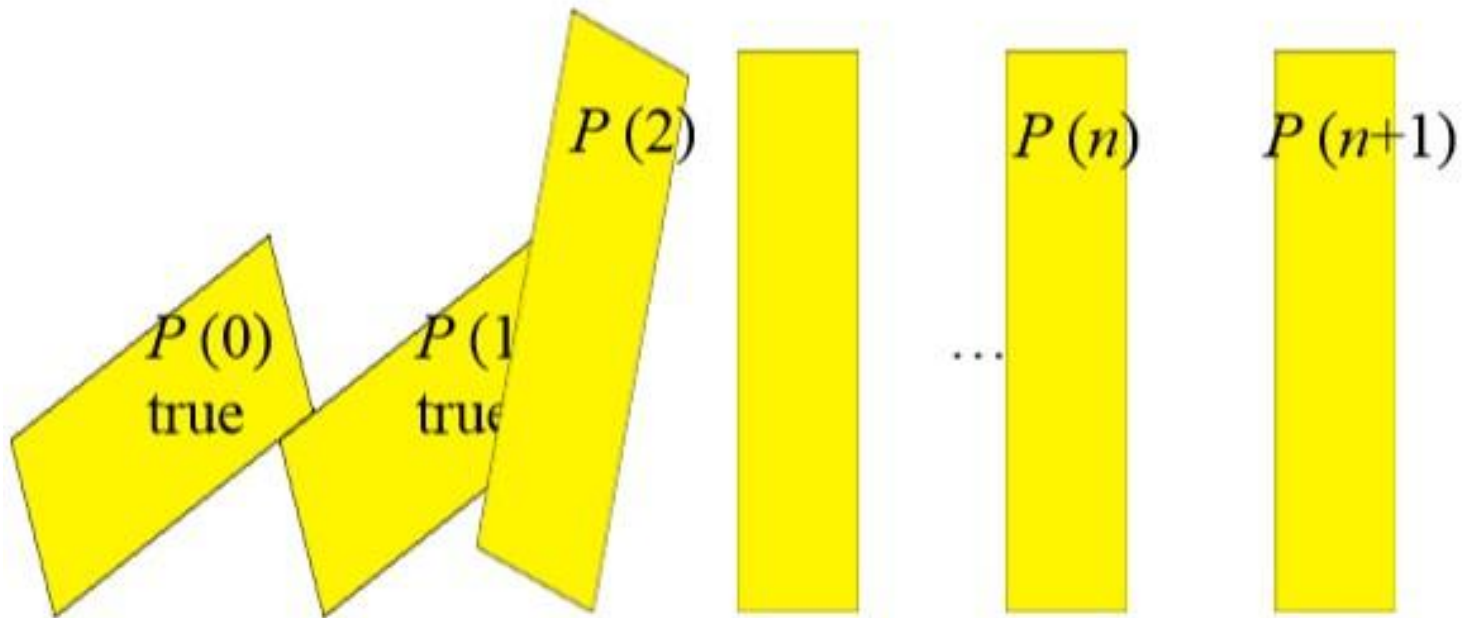
Mathematical Induction

Then we can conclude that all
the dominos fall!



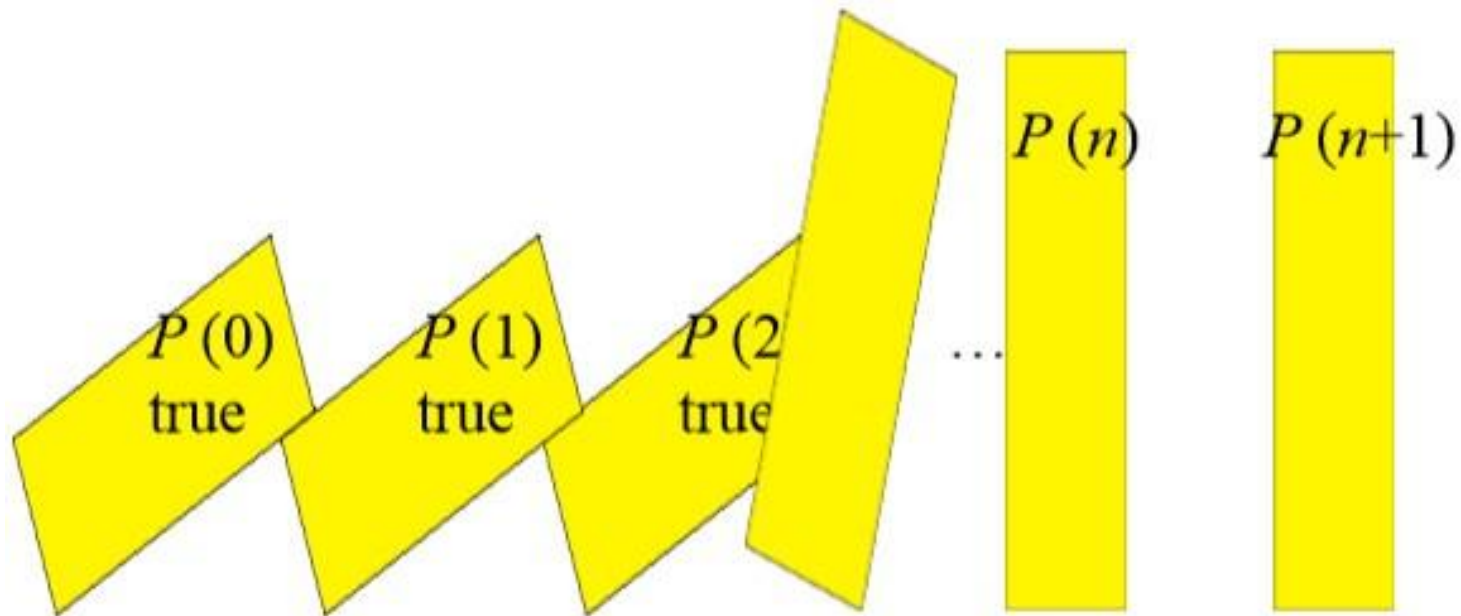
Mathematical Induction

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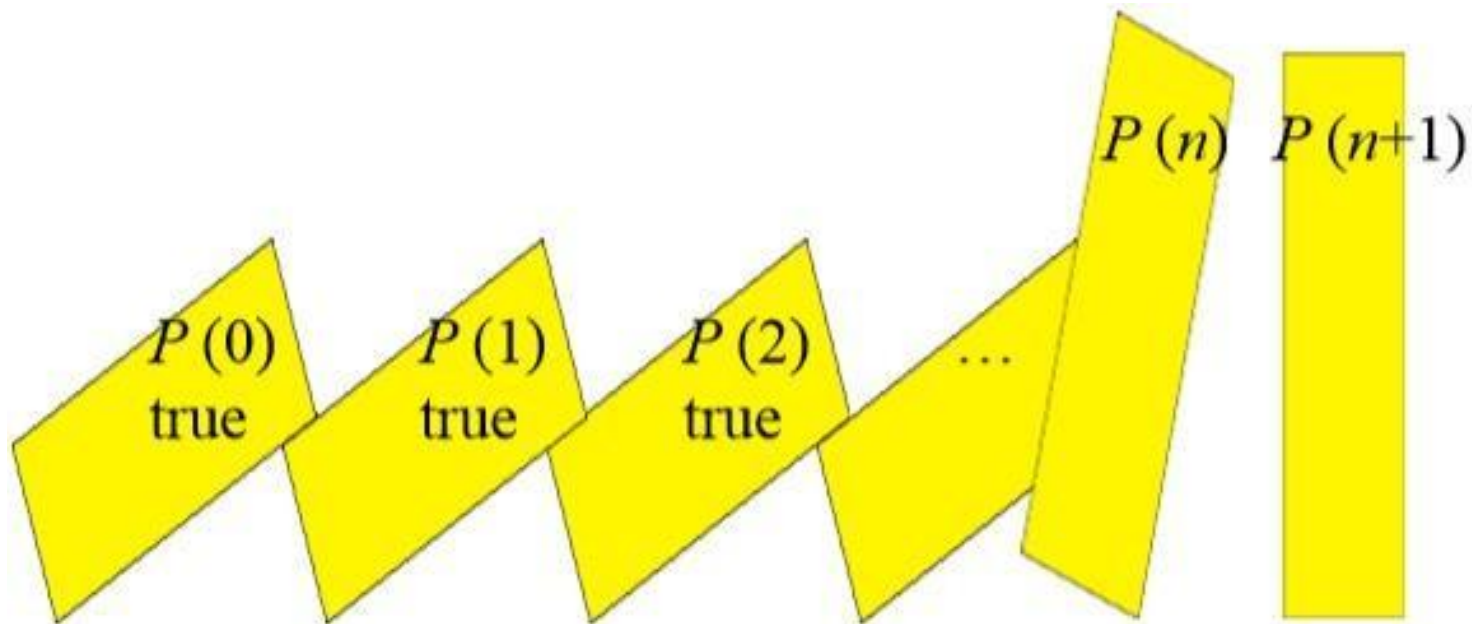
Mathematical Induction

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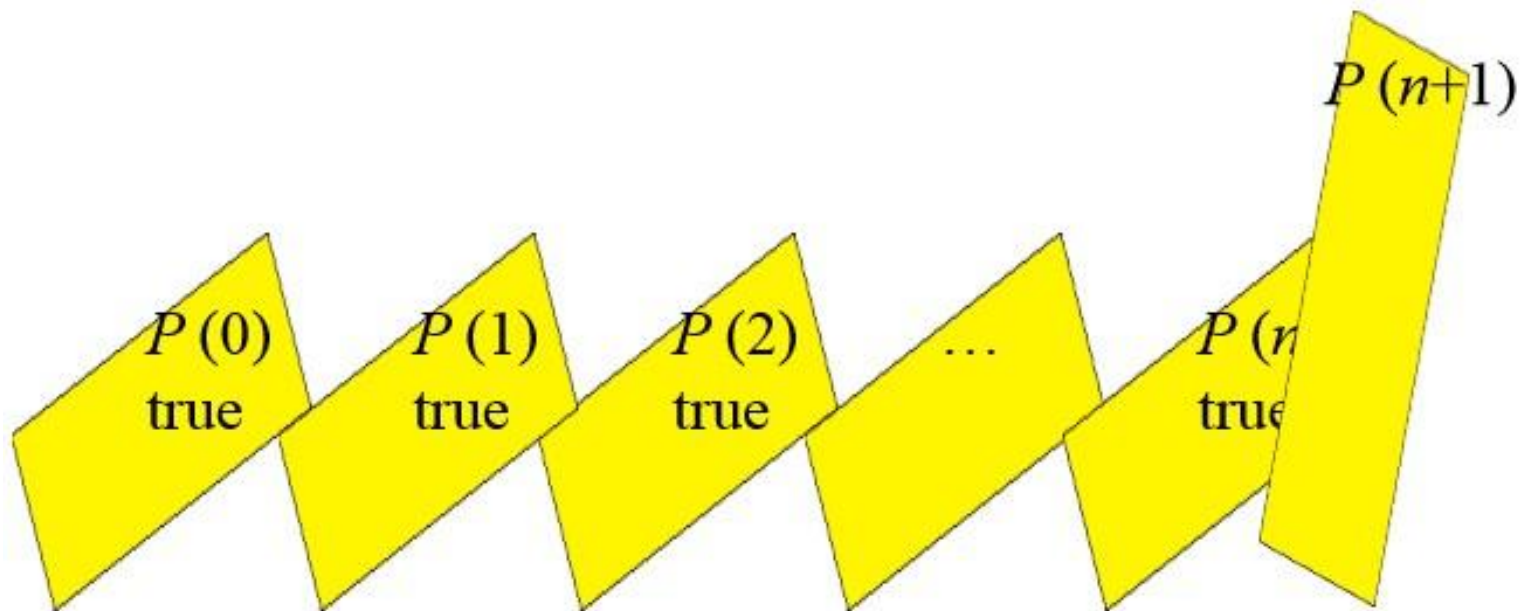
Mathematical Induction

Then we can conclude that all
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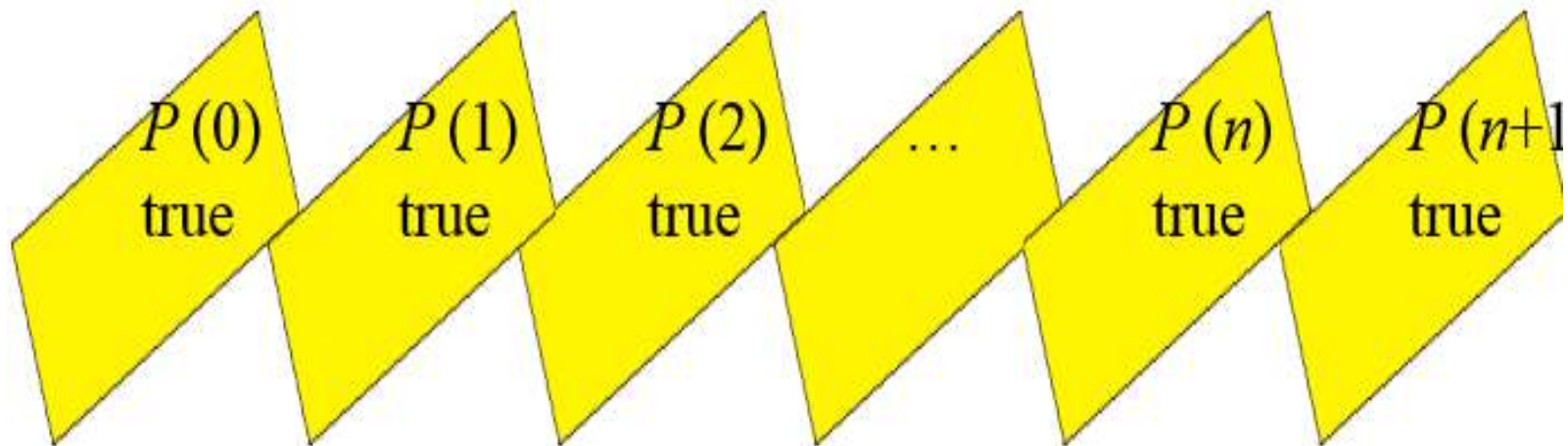
Mathematical Induction

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Mathematical Induction

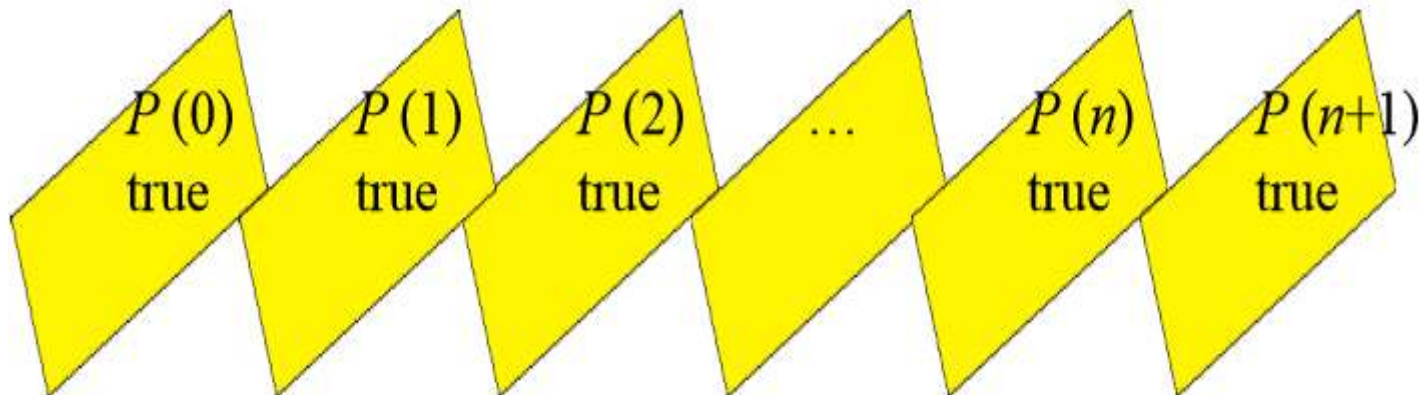
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Mathematical Induction

Principle of Mathematical Induction:

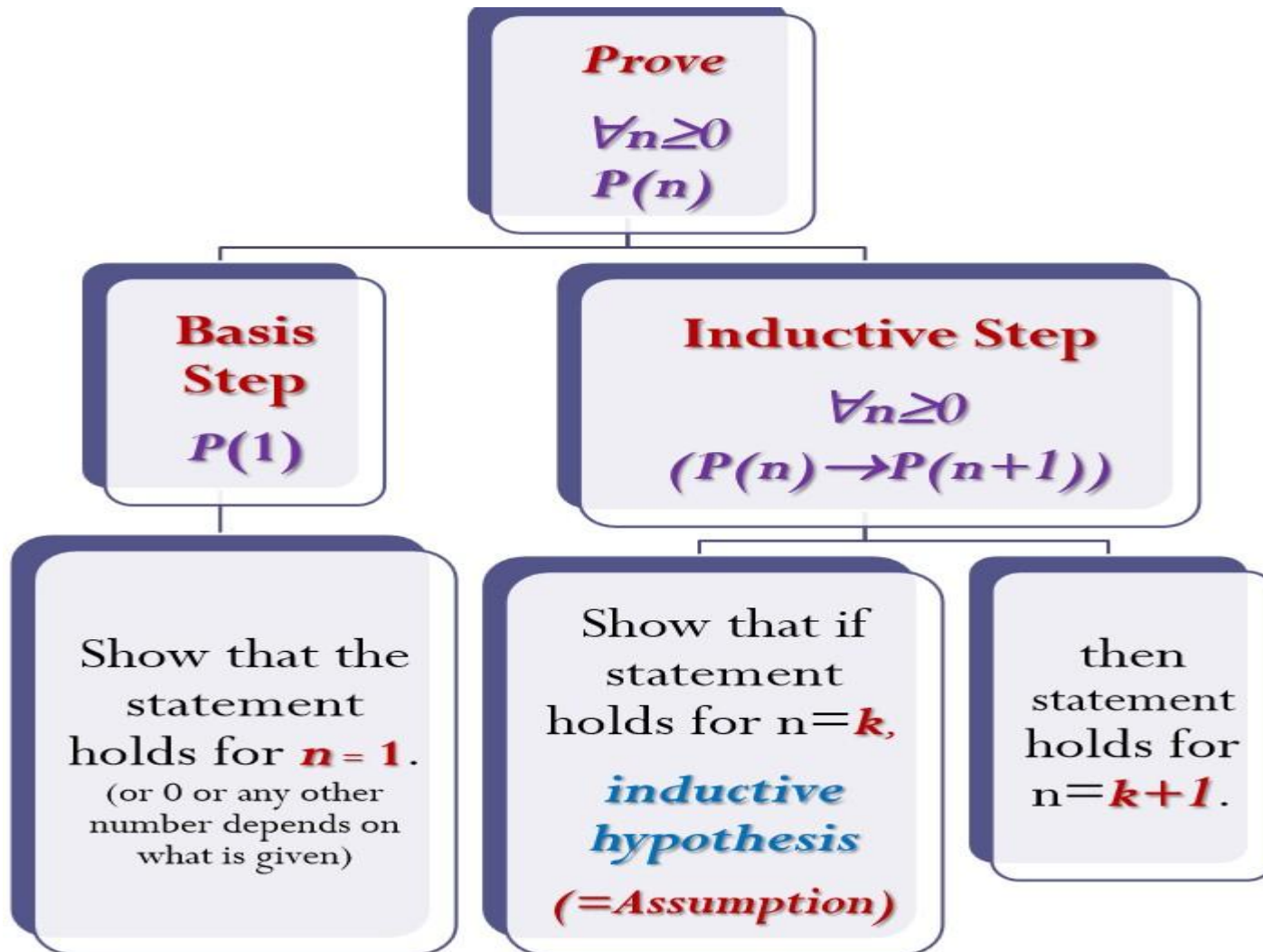
- If:
- 1) [**Basis Step**] $P(0)$ is true
 - 2) [**Induction Step**]
 $\forall n \ P(n) \rightarrow P(n+1)$ is true



$\forall n \ P(n)$ is true

Then: This formalizes what occurred to dominos.

Mathematical Induction



Expressed as a **rule of inference**, this proof technique can be stated as

$$[P(1) \wedge \forall k P(k) \rightarrow P(k+1)] \rightarrow \forall n P(n)$$

Conjecturing and Proving Correct a Summation Formula

Example: Conjecture and prove correct a formula for the sum of the first n positive odd integers. Then prove your conjecture.

Solution: We have: $1=1, 1+3=4, 1+3+5=9, 1+3+5+7=16, 1+3+5+7+9=25.$

- *We can conjecture that the sum of the first n positive odd integers is n^2 ,*

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

- *We prove the conjecture is proved correct with mathematical induction.*

1. **BASIS STEP:** $P(1)$ is true since $1^2 = 1$.

2. **INDUCTIVE STEP:** $P(k) \rightarrow P(k + 1)$ for every positive integer k .

Assume the inductive hypothesis holds and then show that $P(k)$ holds as well.

Inductive Hypothesis: $1 + 3 + 5 + \cdots + (2k - 1) = k^2$

- *So, assuming $P(k)$, it follows that:*

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= [1 + 3 + 5 + \cdots + (2k - 1)] + (2k + 1) \\ &= k^2 + (2k + 1) \text{ (by the inductive hypothesis)} \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

- *Hence, we have shown that $P(k + 1)$ follows from $P(k)$. Therefore the sum of the first n positive odd integers is n^2 .*

Mathematical Induction

Example-p.316

EG: Show that if n is a positive integer, then

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Proof:

Show that if n is a positive integer, then:

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Sol: Basis step: $P(1)$ is true since $1 = 1(1+1)/2$

Inductive step: Assume $P(k)$ holds so that:

$$1 + 2 + \dots + k = k(k+1)/2$$

show that $P(k+1)$ is true, i.e.,

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) &= (k+1)[(k+1)+1]/2 \\ &= (k+1)(k+2)/2 \text{ is true} \end{aligned}$$

Add $(k+1)$ to both sides of the equation of $P(k)$:

$$1 + 2 + \dots + k + (k+1) = k(k+1)/2 + (k+1)$$

$$\begin{aligned} &= k(k+1)/2 + 2(k+1)/2 = (k+1)(k+2)/2, \\ &\text{so } P(k+1) \text{ is true} \end{aligned}$$

Mathematical Induction

Example-p.318

EG: Use mathematical induction to show that

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$$

for all nonnegative integers n .

Mathematical Induction

Example-p.318

EG: Use mathematical induction to prove this formula for the sum of a finite number of terms of a **geometric progression**:

$$\sum_{j=0}^n ar^j = a + ar + ar^2 + \cdots + ar^n = \frac{ar^{n+1} - a}{r - 1} \quad \text{when } r \neq 1,$$

where n is nonnegative integer.

5.3. Recursive Definitions

Recursive Definitions & Induction

Recursive definition and inductive proofs are complement to each others: a recursive definition usually gives rise to natural proofs involving the recursively defined sequence.

This follows from the format of a **recursive definition as**

consisting of two parts:

1. **Initialization** –analogous to induction **base cases**
2. **Recursion** –analogous to induction **step**

In both induction and recursion, the domino analogy is useful.

Recursively Defined Functions

Definition: A *recursive* or *inductive* definition of a function consists of two steps.

1. **BASIS STEP:** *Specify the value of the function at zero.*
2. **RECURSIVE STEP:** *Give a rule for finding its value at an integer from its values at smaller integers.*

Recursively Defined Functions

Basis Step: Specify the value of the function at zero

Example 1: Suppose f is defined by:

Basis step: $f(0) = 3$

Recursive definition: $f(n+1) = 2f(n) + 3$

Find $f(1)$, $f(2)$, $f(3)$ and $f(4)$?

$$f(n+1) = 2f(n) + 3$$

$$n=0, f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$$

$$n=1, f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$$

$$n=2, f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$$

$$n=3, f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$$

Recursive Step:

Give a rule for finding its value at an integer from its values at smaller integers.

Example 2

The recursive definition:

$$f(0) = 1$$

$$f(n) = n.f(n-1)$$

Find $f(5)$?

$$f(n) = n.f(n-1)$$

$$n=1, f(1)=1.f(0) = 1.1 = 1$$

$$n=2, f(2)=2.f(1)= 2 .1= 2$$

$$n=3, f(3)=3.f(2)= 3 .2= 6$$

$$n=4, f(4)=4.f(3)= 4 .6= 24$$

$$n=5, f(5)=5.f(4)= 5 . 24 = 120$$

OR in 1 step like following:

$$\begin{aligned} f(5) &= 5.f(4) = 5.4.f(3) = 5.4.3.f(2) = \\ &5.4.3.2.f(1) = 5.4.3.2.1.f(0) = 5.4.3.2.1.1 = 120 \end{aligned}$$

Example 3

The fibonacci numbers, $f_0, f_1, f_2, \dots$
are

defined by the equations :

$$f_0 = 0, f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2} \quad \text{for } n = 2, 3, 4, \dots$$

Find f_2, f_3 and f_4 ?

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$$f_0=0 \quad , \quad f_1=1$$

$$f_n = f_{n-1} + f_{n-2} \quad \text{for } n=2, 3, 4, \dots$$

Find f_2, f_3 and f_4 ?

$$f_n = f_{n-1} + f_{n-2}$$
$$n=2, \quad f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$n=3, \quad f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$n=4, \quad f_4 = f_3 + f_2 = 2 + 1 = 3$$

Example 4

Given: $f(0)=3$, $f(n+1)=2f(n)$, $n \geq 0$

Find $f(1)$, $f(2)$,..... $f(5)$

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Given: $f(0)=3$, $f(n+1)=2f(n)$, $n \geq 0$

Find $f(1)$, $f(2)$,..... $f(5)$

$$f(n+1)=2f(n)$$

$$n=0, f(1) = 2f(0) = 2*3 = 6$$

$$n=1, f(2) = 2f(1) = 2*6 = 12$$

$$n=2, f(3) = 2f(2) = 2*12 = 24$$

$$n=3, f(4) = 2f(3) = 2*24 = 48$$

$$n=4, f(5) = 2f(4) = 2*48 = 96$$

Example 5

Given: $f(n+1) = f^2(n) - 2f(n) - 2$, $f(0) = 3$,
 $n \geq 0$

Find $f(1)$, $f(2)$

Example 5

Given: $f(n+1)=f^2(n)-2f(n)-2$, $f(0)=3$,
 $n \geq 0$

Find $f(1)$, $f(2)$

$$f(n+1)=f^2(n)-2f(n)-2$$

$$\begin{aligned} n=0, f(1) &= f^2(0) - 2f(0) - 2 \\ &= 3^2 - 2(3) - 2 = 9 - 6 - 2 = 1 \end{aligned}$$

$$\begin{aligned} n=1, f(2) &= f^2(1) - 2f(1) - 2 \\ &= 1^2 - 2(1) - 2 = 1 - 2 - 2 = -3 \end{aligned}$$